Shell

PENG

November 4, 2026

Contents

1 Introduction **1**

2 Shell Syntax **1**
2.1 Quoting . . . 1
2.2 Pipelines . . . 2
2.3 Lists of Commands . . . 2
2.4 Construct . . . 2
2.4.1 Looping . . . 2
2.4.2 Conditional . . . 2
2.4.3 Grouping . . . 2
2.5 Functions . . . 2
2.6 Parameters . . . 3
2.7 Expansion . . . 3
2.8 Redirection . . . 3

3 Array **3**

4 sed **4**

5 awk **4**

6 grep **4**

7 find **4**

A Tools **4**

1 Introduction

`https://www.gnu.org/software/bash/manual/bash.html`

2 Shell Syntax

2.1 Quoting

Definition 2.1 (quoting)**.** *Quoting is used to remove the special meaning of certain characters or words to the shell.*

```
echo -e "\t"
# \n : newline
# \t : horizontal tab
# \r : carriage return
```

2.2 Pipelines

Definition 2.2 (pipeline). *cmd1 output → cmd2 input*

```
cmd1 | cmd2      # stdout
cmd1 |& cmd2     # stderr to stdout
cmd1 2>&1 | cmd2 # stderr to stdout
```

2.3 Lists of Commands

Definition 2.3 (list).

```
cmd1 && cmd2 # And, cmd1 true then cmd2
cmd1 || cmd2 # Or, cmd1 false then cmd2
cmd1; cmd2   # or terminated by newline
cmd1 & cmd2  # cmd1 async
```

2.4 Construct

2.4.1 Looping

Definition 2.4 (looping).

```
for item in item1 item2 ...; do
    cmd
done
```

2.4.2 Conditional

Definition 2.5 (conditional).

```
if [[ cond1 ]]; then
    cmd1
elif [[ cond2 ]]; then
    cmd2
else
    cmd3
fi

# cond:
# ==
# !=
# =~ : unquoted regex
```

2.4.3 Grouping

Definition 2.6 (grouping).

```
( expr )  # subshell env, def var vanish when exit
{ expr; } # ; or newline is required
```

2.5 Functions

Definition 2.7 (function).

```
func () {
    local var = $1
}
```

2.6 Parameters

Definition 2.8 (parameters)**.**

- `${N}` *: position param*
- `$@` *: expands to all pos params,* `$1 $2 ...`
- `$0` *: file or shell name*

2.7 Expansion

Definition 2.9 (brace expansion)**.** *Generate strings sharing a common prefix and suffix.*

```
echo file{1,2} # file1 file2
```

Definition 2.10 (tidle expansion)**.**

```
~ # $HOME
```

Definition 2.11 (parameter expansion)**.**

```
${param}

${param:-default}

${array[@]} # member of array

${!name[@]} # list of array indices
```

Definition 2.12 (command substitution)**.**

```
$(cmd)
`cmd`
```

Definition 2.13 (arithmetic expansion)**.**

```
$((expr)) # integer-only
```

Definition 2.14 (filename expansion)**.**

```
*  # any string
?  # single char
[] # char enclosed between []
*()
?()
+()
!(pattern list) # match except given patterns
```

2.8 Redirection

Definition 2.15 (redirection)**.**

```
ls > file 2>&1   # stdout and stderr to file
ls &> file       # or
pwd >> file 2>&1 # append stdout and stderr to file
pwd &>> file     # or
```

3 Array

Definition 3.1 (array)**.**

```
array=(v1 v2 ...)

for i in ${!array[@]}; do
    echo $array[$i]
done
```

4 sed

5 awk

Definition 5.1 (awk)**.** *The basic function of awk is to search files for lines that contain certain patterns.*

Example 5.1.

```
# % @fig{png}
mapfile -t array < <(awk -F'[{}]' '/^% @fig\{/ { print $2 }' file)
```

6 grep

Definition 6.1 (grep)**.** *Search input file for matches to the patterns.*

7 find

A Tools